

KUTAISI, Georgia

WMO: 373950

Lat: 42.254N

Lon: 42.628E

Elev: 371

StdP: 14.50

Time Zone: 4.00 (E04)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	29.5	31.8	16.9	13.1	38.8	20.2	15.4	40.2	49.0	45.0	43.8	48.7	8.8	90	0.468

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	16.0	95.4	74.1	91.5	73.3	88.4	72.6	78.0	87.9	76.6	85.8	75.3	83.9	18.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.3	134.6	82.6	74.0	128.6	81.5	72.7	123.1	80.2	41.8	88.2	40.4	85.9	39.1	84.1	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 44.1	36.2	31.7	26.4	101.1	3.6	3.1	23.8	103.3	21.7	105.1	19.7	106.8	17.1	109.1		
(5)			23.9	80.4	3.8	1.5	21.1	81.5	18.9	82.4	16.7	83.2	14.0	84.3	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.2	44.5	46.4	50.6	57.9	66.8	73.5	76.6	78.7	72.2	64.0	54.4	47.6	(6)	(7)
	DBStd	13.78	6.65	7.85	8.26	7.84	7.13	5.34	5.29	5.17	6.60	6.51	7.38	7.82	(7)	(8)
	HDD50	633	194	148	95	15	0	0	0	0	0	1	41	139	(8)	(9)
	HDD65	2893	636	520	450	240	65	4	0	0	17	99	322	540	(9)	(10)
	CDD50	4713	23	48	115	251	520	705	825	890	665	434	172	65	(10)	(11)
	CDD65	1498	0	0	4	27	119	259	360	425	232	67	4	1	(11)	(12)
	CDH74	11898	0	0	16	214	992	1950	2835	3884	1686	306	15	0	(12)	(13)
	CDH80	4387	0	0	1	32	322	682	1120	1622	564	44	0	0	(13)	(14)
(14) Wind	WSAvg	10.5	11.5	13.9	12.1	11.1	9.7	8.8	7.6	8.1	9.2	10.4	11.9	11.5	(14)	(15)
(15) Precipitation	PrecAvg	60.7	5.2	4.7	5.3	4.6	4.1	5.7	4.5	4.1	5.5	5.8	5.6	5.6	(15)	(16)
	PrecMax	75.0	12.5	9.8	11.1	6.7	7.4	10.8	8.9	8.3	10.1	13.5	11.1	12.4	(16)	(17)
	PrecMin	47.6	1.5	2.3	2.4	2.0	2.1	3.1	2.4	1.2	0.9	1.7	0.9	1.4	(17)	(18)
	PrecStd	6.5	2.6	1.8	1.9	1.1	1.6	1.8	1.7	1.6	2.4	2.8	3.0	2.5	(18)	(19)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.3	69.4	75.6	83.7	91.4	95.3	99.7	100.6	95.1	85.4	76.3	70.1	(19)	(20)
		MCWB	48.8	54.6	57.6	65.2	69.2	72.5	75.6	74.9	73.2	66.5	59.2	54.2	(20)	(21)
	2%	DB	59.6	64.7	71.2	79.5	87.4	90.4	95.0	95.7	89.8	81.0	71.5	65.5	(21)	(22)
		MCWB	47.1	51.0	55.3	62.6	67.7	72.3	74.8	74.4	71.3	64.9	56.9	52.0	(22)	(23)
	5%	DB	57.1	61.1	67.1	76.0	83.5	86.9	89.8	91.3	85.7	77.7	67.6	61.4	(23)	(24)
		MCWB	46.0	49.2	53.6	60.6	66.6	71.3	73.8	74.5	69.7	63.6	55.3	49.6	(24)	(25)
	10%	DB	54.3	57.6	63.5	71.9	79.5	83.7	86.0	88.1	82.5	74.4	64.2	58.1	(25)	(26)
		MCWB	44.4	47.0	51.8	58.2	65.1	70.3	73.6	73.9	68.6	62.2	53.7	47.9	(26)	(27)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.1	55.8	59.4	67.0	71.1	77.6	79.7	79.8	76.6	68.3	60.3	56.9	(27)	(28)
		MCDB	58.2	66.3	72.9	81.1	85.8	87.4	90.1	90.1	88.3	79.3	71.8	64.6	(28)	(29)
	2%	WB	48.5	52.0	56.5	63.6	69.3	75.2	77.8	78.3	74.1	66.5	58.4	53.2	(29)	(30)
		MCDB	55.8	61.9	68.7	78.1	83.2	85.4	88.0	88.0	83.9	76.7	67.5	61.1	(30)	(31)
	5%	WB	47.1	49.9	54.4	61.2	67.7	73.4	76.3	77.0	72.2	65.2	56.8	51.1	(31)	(32)
		MCDB	54.4	59.2	65.5	74.4	80.8	83.1	85.7	85.9	81.5	74.8	64.9	59.0	(32)	(33)
	10%	WB	45.7	47.9	52.4	59.0	65.9	71.7	74.8	75.7	70.5	63.6	54.9	49.4	(33)	(34)
		MCDB	52.8	56.1	61.7	70.6	77.6	80.7	82.9	84.0	79.2	72.3	62.3	56.8	(34)	(35)
(35) Mean Daily Temperature Range	5% DB	MDBR	10.6	12.0	13.4	16.5	16.3	15.4	13.9	16.0	15.9	14.1	12.1	10.7	(35)	(36)
		MCDBR	14.2	18.0	19.7	23.7	23.1	21.3	21.0	21.5	20.7	19.1	17.7	15.1	(36)	(37)
	5% WB	MCWBR	8.2	10.2	10.7	12.0	10.6	9.0	8.3	8.6	9.0	9.5	9.8	8.6	(37)	(38)
		MCWBR	12.3	17.3	18.4	22.4	21.3	18.9	17.4	18.9	18.9	17.0	15.6	13.2	(38)	(39)
(39) Clear-Sky Solar Irradiance	taub		0.329	0.362	0.415	0.465	0.435	0.424	0.451	0.454	0.401	0.390	0.346	0.323	(39)	(40)
		taud	2.303	2.195	2.110	2.057	2.237	2.330	2.279	2.268	2.397	2.373	2.435	2.372	(40)	(41)
	Ebn at Noon		253	261	260	256	268	271	261	256	264	251	247	245	(41)	(42)
		Edh at Noon	28	36	45	51	43	40	42	41	33	30	24	24	(42)	(43)
(43) All-Sky Solar Radiation	RadAvg		517	784	1044	1413	1682	1861	1841	1727	1391	941	630	457	(43)	(44)
	RadStd		73	95	94	167	170	144	173	131	134	105	82	47	(44)	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (14 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page